Performance Testing

Date	3 July, 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Manual Testing (Web Browser)
Type of Testing	Functional Performance Testing
Target Module	Credit Card Approval Prediction (Flask Prediction Module)
Test Environment	Local System (Flask Development Server)
Test Date	3 July 2026

Step 2: Test Scenarios

S.No	**Test Scenario / Description**	**No. of virtual Users**	**Duration (sec)**	**Expected Outcome**
1	Submit a valid credit card application	1	10	Prediction generated Prediction result is displayed successfully
2	Submit multiple prediction requests	5	30	Application responds correctly without errors
3	Peak load testing	200	180	responsive and maintains acceptable performance

Step 3: Performance Test Results

S.No	**Metric**	**Target Value**	**Actual Value**	**Status (pass/fail)**	**Remarks**
1	Response Time (Avg)	<2seconds	1.4 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.63 sec	Pass	Within target
3	Throughput (Req/sec)	stable	8 req/sec	Pass	Suitable for small-scale testing
4	Error Rate	<1%	0%	Pass	No errors during testing
5	CPU Utilization	<80%	45%	Pass	Normalresource usage
6	Memory Utilization	<80%	52%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

- The application responded quickly to user requests.
- Credit card predictions were generated successfully.
- No critical errors were observed during testing.
- The system remained stable throughout the test

Bottlenecks Identified

- Minor delay during the initial model loading; subsequent predictions were faster.
- Performance depends on the local system's hardware and available memory.
- No critical performance bottlenecks were observed.

Optimization Steps Taken

- Optimized data preprocessing for faster predictions.
- Used a trained machine learning model for efficient prediction.
- Improved input validation to reduce errors.
- Organized the code into reusable modules for better maintainability.

Step 5: Screenshots / Evidence